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Solutions Architect with 7+ years at AWS serving as a trusted technical advisor to enterprise and public-sector customers across India. Proven track record of translating complex business challenges into scalable cloud and generative AI solutions, while driving multi-million dollar deals, helping customers migrate and modernize on cloud, delivering measurable cost optimization and guiding stakeholders from C-suite to engineering teams. AWS Direct Connect SME with deep expertise in networking, infrastructure, cloud native and agentic AI architectures.

CORE COMPETENCIES / KEY SKILLS

- **Certifications:** AWS Certified Solutions Architect - Professional & Associate | AWS Certified Advanced Networking - Speciality | AWS Certified AI Practitioner | AWS Certified Speaker Associate
- **Architecture Domains:** Serverless, Microservices/Containerized workloads, Security & Compliance, Data Management (Data Lake/Warehouse), Data analytics, Cloud AI, Disaster Recovery, Multi-Account/Multi-Region Design, Cost Optimization
- **Technical Domains:** Network Architecture, Cost Modelling & TCO Analysis, Infrastructure as Code, Well-Architected Reviews, Onprem/cloud to cloud migrations, Application modernization, Infrastructure modernization
- **AI/ML:** LLMs, RAG, Multimodal Vision, Agentic AI workflows, multi-agent systems, AI evaluation and orchestrator frameworks
- **Security:** Identity and Access Management, Encryption, Access Control, Zero-Trust, Data residency, Secure and Responsible AI

PROFESSIONAL EXPERIENCE

DEC 2024 – PRESENT

SOLUTIONS ARCHITECT, AMAZON WEB SERVICES (AWS)

- **UNICEF - RapidPro Platform Automation:** Designed and deployed scalable RapidPro infrastructure on AWS for UNICEF India with fully automated single-click provisioning via CloudFormation, targeting adoption across multiple government ministries. Cut implementation time and secured a technical relationship positioning AWS for a **20x monthly billing pipeline expansion**.
- **ISHA Foundation - High-Scale Event and Application Launch Support:** Architected and validated the scaling strategy across ECS, EC2, and Load Balancers for ISHA's Mahashivaratri 2025 event, handling a traffic surge of **1.4M app downloads in 12 hours with zero downtime**. Provided emergency support for a last-minute campaign to 7M users by securing a temporary SES increase to 500 TPS over a 4-hour window.
- **Government Healthcare Agency - Strategic Cost Optimization Analysis:** Ran a Bill of Quantities analysis and technical validation for a major Government Healthcare Agency's workload renewal, benchmarking AWS against two competing CSPs. A tailored cost-optimization strategy demonstrated **~80% lower projected running cost** than rival CSPs, positioning AWS as the cost-to-performance leader in a competitive RFP and driving a high-value government sales motion.
- **Coal India Limited (CIL) - SAP RISE Migration Network Architecture:** Led network architecture design and deployment for the strategic SAP ECC-to-RISE migration on AWS, a **\$10M (6-year TCO) win**. Implemented Site-to-Site VPN for the one-time bulk data transfer, then provisioned resilient, low-latency connectivity for SAP users via multiple dedicated Direct Connect links. This foundational work secured the deal and **unlocked a further \$18M in cloud opportunity**.
- **Echo India - Cost-Optimized Multi-Agent RAG Solution:** With the Cloud Innovation Center team, designed an autonomous multi-agent RAG chatbot using Strands SDK and Amazon Bedrock. An S3 vector store delivered **90% cost savings** over traditional vector databases (under \$2,000/month at peak), and an evaluation framework reached **>90% accuracy** across two domains - driving the customer's decision to expand to a full 73-disease implementation.
- **Ministry of Women & Child Development (MWCD):**
 - **PMMVY Well-Architected Review & Architecture Optimization:** Delivered a Well-Architected review and target architecture for the production PMMVYSoft MIS (multi-AZ maternity-benefit DBT platform under Mission Shakti), eliminating single points of failure across web, mobile, API, file server, and UIDAI/PFMS tiers. Converted a pseudo-single-active 3-node web tier to fully active/multi-AZ via ElastiCache (Valkey) session externalization and Auto Scaling; right-sized the Aurora PostgreSQL read tier with Aurora Auto Scaling, RDS Proxy pooling, and read/write splitting; consolidated 3x redundant NLB/WAF/ALB stacks into a single CloudFront + WAF + ALB edge; and replaced a

~10TB single-instance file server with EFS and lifecycle-tiered S3 - strengthening reliability and cost efficiency on a mission-critical payment system.

- **AI-Powered Nutritional Analysis for Poshan Tracker:** Built a POC using Amazon Bedrock (Claude) multimodal vision to analyze Anganwadi meal photos - identifying food items, estimating portions, and scoring nutritional compliance against ICDS norms. The fully serverless stack (Bedrock, Lambda, S3, DynamoDB) delivers **zero infra management and elastic scale to 12L+ daily images across 14L centers serving ~9Cr beneficiaries**. Success opened follow-on AI image-analysis discussions, including AI-based child height/weight estimation (in prospect stage).
- **PMMVY Disaster Recovery Architecture:** Revised a stalled DR proposal for the PMMVY workload, re-architecting the DR-region database from Aurora Standard to Aurora Headless for a **>75% DR cost reduction while preserving RPO/RTO**. Broke months of deadlock to secure MWCD's acceptance and formal approval to begin implementation.
- **Poshan Tracker Cost Optimization Drive:** Led an end-to-end cost review of the Poshan Tracker production workload, delivering **>30% monthly billing reduction**. Largest gains came from re-engineering NAT Gateway traffic - provisioning interface and gateway VPC endpoints to route internal traffic directly to AWS services instead of the costlier NAT path.
- **Quicko - Multi-Account Network Architecture Advisory:** As AWS Network Specialist, reviewed Quicko's multi-control-tower network design ahead of org-wide rollout. Recommended a centralized Transit Gateway hub-and-spoke (shared via RAM) over subnet-sharing to preserve per-account isolation and avoid CIDR exhaustion across 15–20 workloads per control tower, with centralized egress/inspection and per-control-tower NAT Gateways for throughput headroom and per-BU cost attribution. For 1B+ internal API calls/month between sister entities (Quicko ↔ Sandbox), kept traffic on the AWS backbone via TGW peering (or PrivateLink for non-org consumers) and closed the public-DNS gap with a Route 53 private hosted zone - resolving the same domain to private IPs internally while public resolution served external users.
- **Federal Accounts – Public Sector Networking:** Contributed to major public-sector opportunities including NIC's AWS Outposts migration (Meghraj 2.0) and SAP RISE migrations for India PSUs (GAIL, CIL, BL).
- **Speaking & Enablement:** Represented AWS at India Impact AI Summit, UP AI Conclave (presented before the UP CM), New Delhi Symposium, NHA NHCC Hackathon, and Arohana. Delivered customer sessions including EKS enablement for Dhanush Healthcare, Networking immersion day for Uplight, Network connectivity session for EDF, CDN and edge services for NIC, and a full-day AI workshop at AIIMS New Delhi (avg CSAT ~4.5).

JUL 2019 – NOV 2024

CLOUD SUPPORT ENGINEER, AMAZON WEB SERVICES (AWS)

- **Team Enablement & Quality:** Conducted 50+ interviews, delivered 30+ trainings, and reviewed 100+ support cases (quality, technical, and SME evidence reviews).
- **AWS Enabler session 2023:** Delivered "Advanced Computer Networking" session to 3,000+ final-year students.
- **Bootcamp:** Organised a 6-day Direct Connect SME bootcamp in North America time-zone.
- **Architectural reviews:** Delivered end-to-end Direct Connect reviews — best practices, limits, performance testing, and redundant setup design [Pepsico, Paytm, Singapore Press Holdings].
- **Trainings:** Delivered a Site-to-Site VPN best-practices session with live demos [SAP], and Direct Connect / S2S VPN / Transit Gateway sessions in a 4-day networking bootcamp for 160+ participants [Accenture].
- **Well-architected designs:** Designed network architectures spanning VPC peering, TGW, VGW, VPN, and DX, comparing patterns on complexity, limits, scalability, and cost [Godigit, Freemium]; advised on static/BGP routing in DX/TGW and SD-WAN across multiple on-prem sites and AWS regions [Maxeon].
- **Escalations:** Owned major escalations and multi-week, multi-stakeholder troubleshooting [Firstsource, Philip Futures], plus E2M and Dive-and-Save recoveries for languished cases [Valuefirst].

PROJECTS

- **Zero-Trust Dynamic Access Control at Scale:**
Architected a scalable, multi-account zero-trust network access solution using fully managed AWS services (Client VPN, Transit Gateway, Network Firewall, Lambda, DynamoDB) to enforce per-user, per-port access control with automated session lifecycle and full audit trail. Delivered a highly available, event-driven design supporting 400 users across 20 accounts at ~\$5/user/month - 50–75% cheaper than third-party ZTNA platforms - while eliminating operational overhead of managing per-account security rules.
- **Hub-and-Spoke Multi-Account Network Connectivity:**
Designed a highly available hub-and-spoke architecture enabling multi-account, multi-region workloads to route through a single central service VPC for internet egress, on-premises connectivity, shared VPC endpoints (SSM, S3, DDB), and Route 53 resolver access - consolidating networking resources and reducing per-account complexity.
- **Centralized Traffic Inspection for Egress and Ingress Workloads:**
Deployed a dual-layer inspection architecture: (1) AWS Network Firewall with Gateway Load Balancer for egress traffic intrusion prevention, identity-aware filtering, and TLS/HTTPS inspection; (2) Network Load Balancer with destination NAT through Checkpoint firewalls for deep packet inspection of ingress traffic to EKS workloads.